

Public/Private/Hybrid: ye decide karte hain infrastructure kahaan hai aur kiske control mein hai. (Cloud Model)

IaaS/PaaS/SaaS: ye decide karte hain ki us infrastructure ko manage kaun karega aur tum kitni abstraction use kar rahe ho. (Service Model)

Pehle On-Premise samajho : Tumhari company khud sab kuch kharidti hai:

- Company Office
 - └— Physical Servers
 - └— Networking Equipment
 - └— Cooling Systems
 - └— Power Backup
 - └— IT Team

Sab company ka. Ye **On-Premise (On-Prem)** hai.

Public Cloud : Tum kuch kharidte nahi ho. Sb cloud me hota hai. (Most startups and modern companies)

- Frontend → Vercel/ S3 + CloudFront
- Backend → AWS ECS / EC2 / Lambda
- Database → RDS
- Storage → S3
- Monitoring → Datadog

AWS ka infrastructure use karte ho. AWS maintain karta hai. Tum pay-as-you-go pay karte ho.

Examples: AWS, Azure, Google Cloud

Private Cloud : Ab yahan log confuse ho jate hain.

Private Cloud ka matlab: **Cloud jaisi capabilities but dedicated to a single organization.**

Ye do tarah se ho sakta hai.

Option 1: Company ke khud ke data center mein

- Company Data Center
 - └— Servers
 - └— Virtualization
 - └— Self-service portal

Company khud cloud jaisa environment banati hai.

Option 2: AWS dedicated infrastructure

AWS bolta hai:Ye poora hardware sirf tumhara. Shared nahi.

OS/Other management: It will depend on what managed service you are taking, like EC2 hua toh khud kro, Beanstalk hua toh AWS krega management like monitor and scaling

Hybrid Cloud: (Most common in large enterprises)

Yahan dono hote hain. **On-Premise infrastructure + Public Cloud working together.**

Company Data Center ---- > (VPN or Direct Connected to) --- > AWS / Azure / GCP

Example:Payroll System will be on ---- > On-Prem

Customer Facing Website will be on --- > AWS

Used mainly in: Banks, Insurance companies, Healthcare

AWS EC2 ---- > Public Cloud + IaaS

AWS Elastic Beanstalk --- > Public Cloud + PaaS

Company VMware Setup ---- > Private Cloud + IaaS



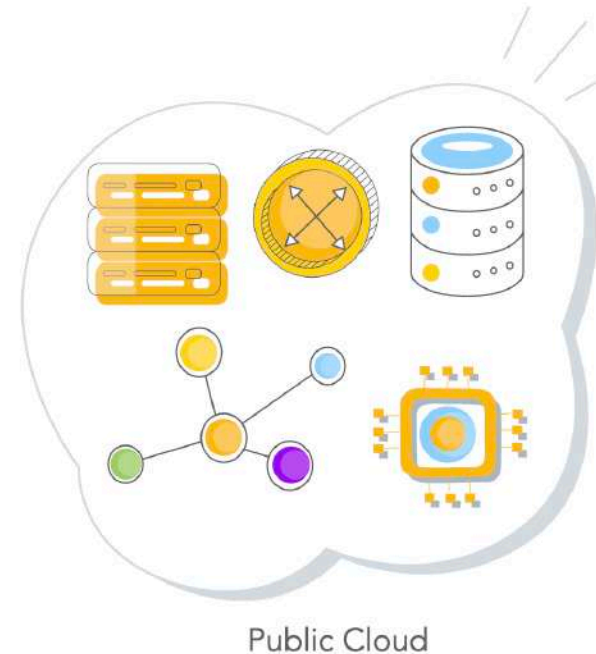
Cloud Computing Model: Public Cloud

Public Cloud: Host everything on the cloud providers infrastructure

- **No Maintenance:** You do not need your own data center anymore
- **No Capital Expense (No CapEx):** Pay as you go (OpEx)
- **Unlimited Scalability:** Add resources when needed

Shared Responsibility:

- **Cloud Provider:** Owns hardware, handles failures & security of the data center
- **You:** Focus on your application, configuration and data



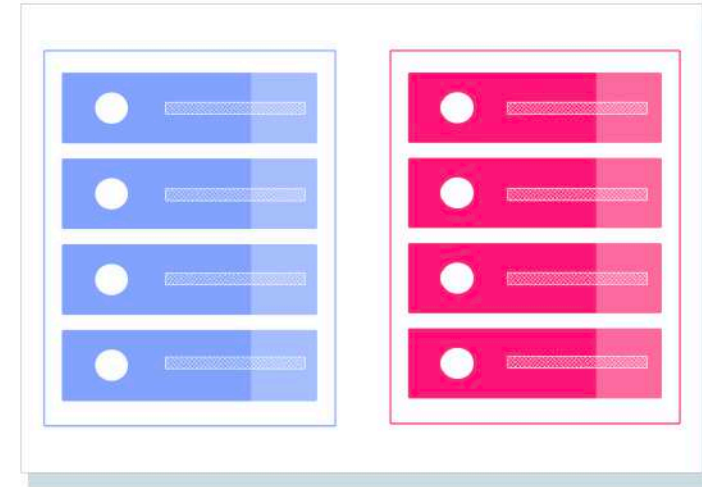


Cloud Computing Model: Private Cloud

Private Cloud: You host everything in your own data center

- **Complete Control:** Over security and hardware
- **High CapEx:** Requires huge investment ahead of time (Hardware, Space, Cooling, ..)
- **Maintenance:** You pay for staff to manage the infrastructure

Slow Scaling: Adding capacity takes weeks or months (purchasing hardware)



Data Center (Private Cloud)



Cloud Computing Model: Hybrid Cloud

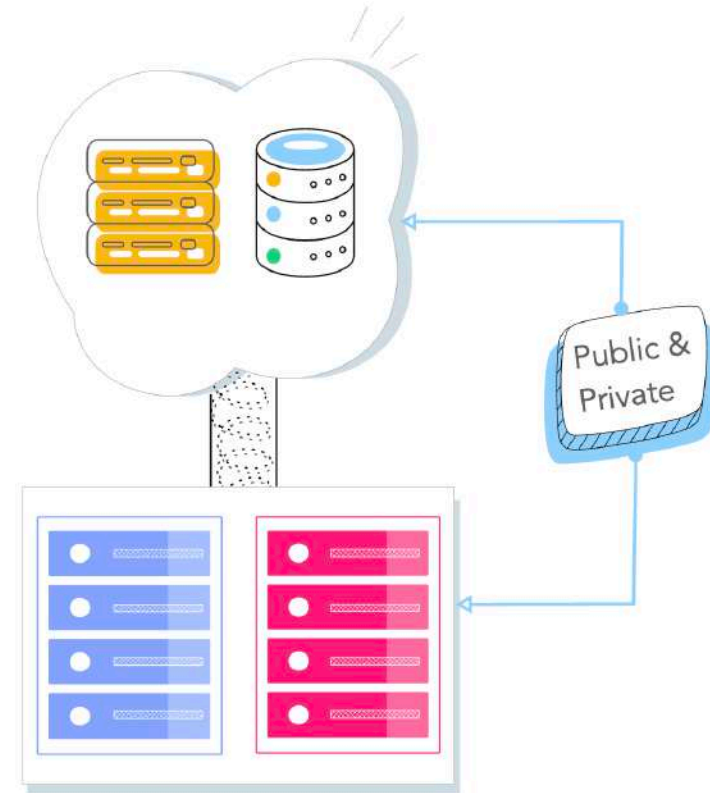
Hybrid Cloud: Public + Private Clouds

- **Example:** Connect an on-prem application to a cloud database
- **Best of Both Worlds:** Use public for scale, private for compliance
- **Flexibility:** Go on-premises or cloud based on specific requirement

Connectivity: Requires secure connection between on premises and cloud

Trade-off: Increases complexity

- Needs management of public and private cloud





Cloud Computing Model: Multi Cloud

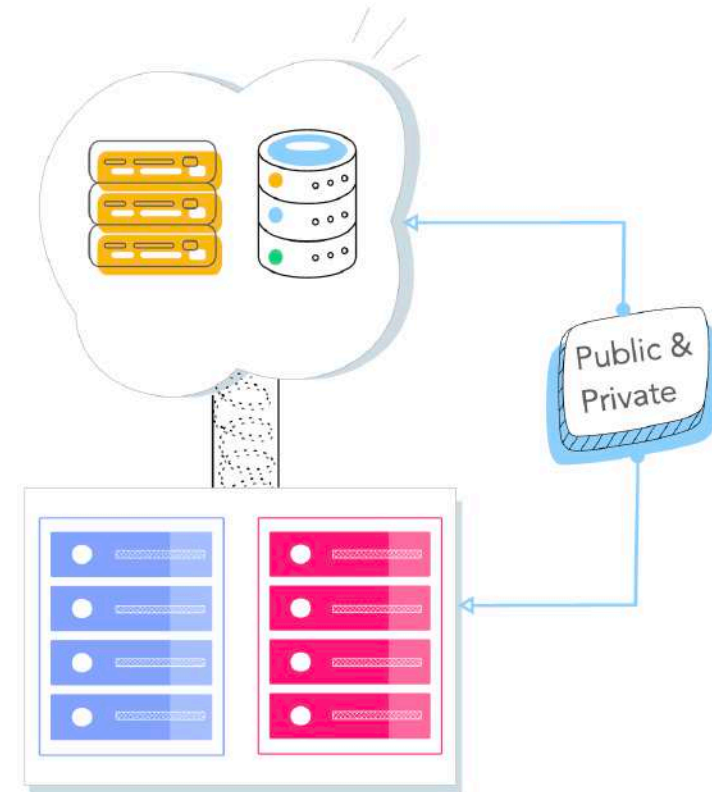
Multi-Cloud: Using multiple public cloud providers (e.g., Google Cloud + AWS + Azure) with/without on-premise infrastructure

Why do this?:

- **Avoid Vendor Lock-in:** Don't depend on a single provider
- **Best of Breed:** You get a lot of choice
- **Reliability:** Failover to another cloud in case of failure

Challenge: Very high complexity

- Skills, Networking, security and management





Cloud Deployment Models - Scenarios

Scenario	Solution
Startup: You want to launch quickly with zero upfront cost and no hardware management	Public Cloud
Banks/Government: You have strict laws requiring data to never leave your physical building	Private Cloud
Modernization: You want to keep your secure mainframe database on-premise but build a new mobile app in the cloud	Hybrid Cloud
Redundancy: You want to ensure your app stays online even if Your Cloud Provider has a global outage	Hybrid-Cloud or Multi-Cloud (GCP + AWS + Azure)
Burst Traffic: You host an e-commerce site on-premise but need extra servers only for Black Friday	Hybrid Cloud (Cloud Bursting)